



Name: _____

Teacher: _____

Class: _____

FORT STREET HIGH SCHOOL

2010

HIGHER SCHOOL CERTIFICATE COURSE

ASSESSMENT TASK 2

Mathematics Extension 1

TIME ALLOWED: 1½ HOURS

Outcomes Assessed	Questions	Marks
Manipulates algebraic expressions to solve problems involving inverse functions	5, 6	
Synthesises mathematical solutions to harder problems and communicates them in appropriate form	1, 2, 3, 4	

Question	1	2	3	4	5	6	Total	%
Marks	/9	/11	/8	/12	/10	/10	/60	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started in a new booklet

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1 (9 marks) Use a SEPARATE writing booklet

- a) Prove the identity $\frac{1 - \tan^2 A}{1 + \tan^2 A} = \cos 2A$. [2]
- b) The point $P(-3, 8)$ divides the interval AB externally in the ratio $k : 1$. If A is the point $(6, -4)$ and B is the point $(0, 4)$, find the value of k . [2]
- c) Solve for x : $\left| \frac{5}{x} + 3 \right| > 3$. [3]
- d) When the polynomial function $f(x)$ is divided by $x^2 - 16$, the remainder is $3x - 1$. [2]
What is the remainder when $f(x)$ is divided by $x - 4$?

Question 2 (11 marks) Use a SEPARATE writing booklet

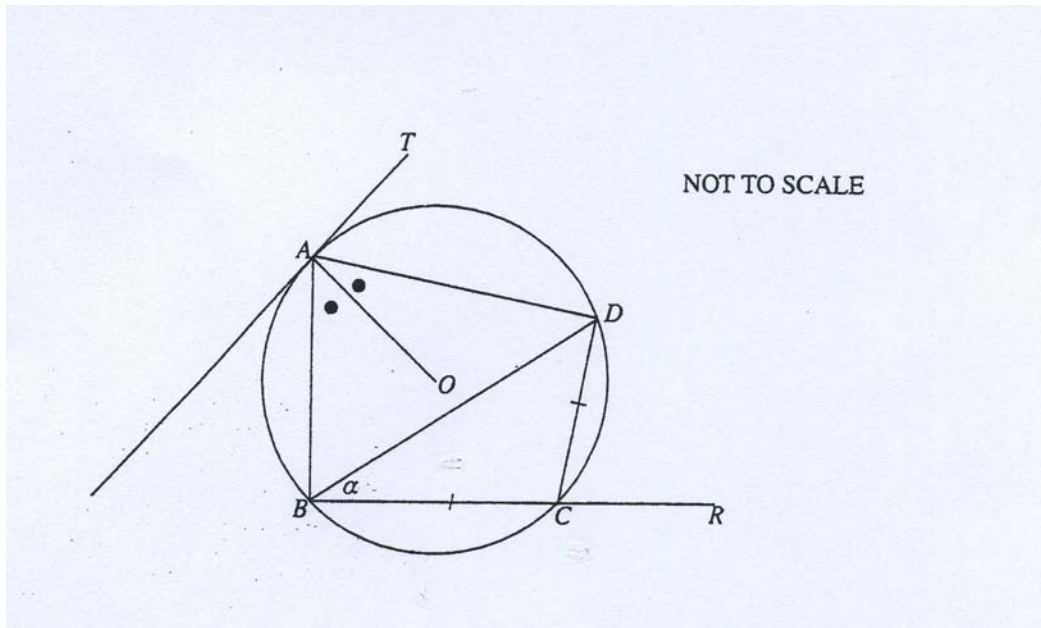
- a) Find the exact value of $\sin 105^\circ$. [2]
- b) Find all values of θ for which $\tan \theta = 3 \cot \theta$. (give your answer in radians). [2]
- c) Show that $\cos(A + B) + \cos(A - B) = 2 \cos A \cos B$. Hence or otherwise, evaluate $\int_0^{\frac{\pi}{6}} 2 \cos 3x \cos 2x \, dx$. [3]
- d) Express $\sin x$ and $\cos x$ in terms of $t = \tan \frac{x}{2}$ and hence find the solution of $5 \cos x - 2 \sin x = 2$ where $0 \leq x \leq 2\pi$. [4]

Question 3 (8 marks) Use a SEPARATE writing booklet

a) Evaluate $\lim_{\theta \rightarrow 0} \frac{\theta + \sin 2\theta}{3\theta}$

[3]

b)



Points A, B, C and D lie on a circle centre O. The line TA is a tangent to the circle at A, and BC is produced to R. The interval OA bisects $\angle BAD$, and $BC = CD$. The size of $\angle DBC$ is α .

Copy or trace the diagram into your writing booklet.

i) Explain why $\angle DCR = 2\alpha$.

[1]

ii) Show that $\angle OAD = \alpha$.

[2]

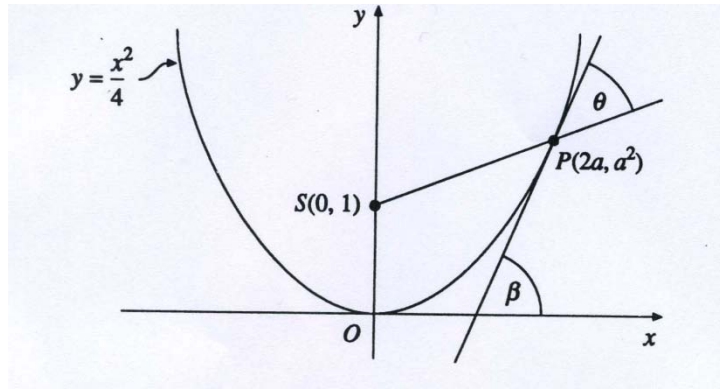
iii) Prove that $\angle ABC$ is a right angle.

[2]

Question 4 (12 marks) Use a SEPARATE writing booklet

- a) For $P(x) = x^3 - 6x^2 + ax - 4$ and $a > 0$ given that all the roots of $P(x) = 0$ are real and positive, and that one of the roots is the product of the other two, show $a = 10$. [4]

b)



Let $P(2a, a^2)$ be a point on the parabola

$$y = \frac{x^2}{4},$$

and let S be the point $(0, 1)$. The tangent to the parabola at P makes an angle of β with the x axis. The angle between SP and the tangent is θ . Assume that $a > 0$, as indicated.

- i) Show that $\tan \beta = a$. [1]
- ii) Show that the gradient of SP is $\frac{1}{2}\left(a - \frac{1}{a}\right)$. [1]
- iii) Show that $\tan \theta = \frac{1}{a}$. [2]
- iv) Hence find the value of $\theta + \beta$. [2]
- v) Find the coordinates of P if $\theta = \beta$. [2]

Question 5 (10 marks) Use a SEPARATE writing booklet

a) Sketch $y = 4\sin^{-1}(3x)$ clearly showing the domain and range and any intercepts. [2]

b) Consider the function $f(x) = x(x-2)^2$.

(i) Sketch the function $f(x)$ given that stationary points exist at $x = \frac{2}{3}$ and $x = 2$. [2]

(ii) Now consider $f(x) = x(x-2)^2$, where $x \leq a$ (a is a constant). Find the value of a for which $f^{-1}(x)$ can exist. Give a reason for your answer. [2]

(iii) State the domain of $f^{-1}(x)$. [1]

c) Find the exact values of x and y which satisfy the simultaneous equations

$$\sin^{-1} x - \cos^{-1} y = \frac{\pi}{12} \quad \text{and} \quad \sin^{-1} y + \cos^{-1} x = \frac{5\pi}{12} \quad [3]$$

Question 6 (10 marks) Use a SEPARATE writing booklet

(a) Find the primitive function of $\frac{1}{\sqrt{4-x^2}}$ [1]

(b) Differentiate $\cos^{-1}(3-2x)$, giving your answer in the simplest factored form. [3]

(c) Evaluate $\int_0^{\frac{1}{\sqrt{3}}} \frac{dx}{1+9x^2}$ giving your answer in exact form. [2]

(d) The region in the first quadrant bounded by the curve $y = 3 \tan^{-1} x$ and the y-axis between $y = 0$ and $y = \pi$ is rotated through one complete revolution about the y-axis. Find the volume of the solid of revolution in exact form. [4]



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$$a) \frac{1 - \tan^2 A}{1 + \tan^2 A}$$

$$1 + \tan^2 A$$

$$= \frac{\cos^2 A - \sin^2 A}{\cos^2 A} \div \frac{\cos^2 A + \sin^2 A}{\cos^2 A} \quad \checkmark$$

$$= \frac{\cos^2 A - \sin^2 A}{\cos^2 A + \sin^2 A} \quad \checkmark$$

$$= \cos^2 A - \sin^2 A$$

$$= \cos 2A$$

OR

$$1 - \tan^2 A$$

$$1 + \tan^2 A$$

$$= \frac{\cos^2 A - \sin^2 A}{\cos^2 A} \times \frac{1}{\sec^2 A} \quad \checkmark$$

$$= \frac{\cos^2 A - \sin^2 A}{\cancel{\cos^2 A}} \times \cancel{\cos^2 A}$$

$$= \cos^2 A - \sin^2 A \quad \checkmark$$

$$= \cos 2A$$

Comments

1 for writing
expression in terms
of $\cos A$, $\sin A$

1 for knowing this
is equal to
 $\cos 2A$.

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b) $P(-3, 8)$ $A(6, -4)$ $B(0, 4)$

$k: -1$ (external division)

$$x = \frac{kx_2 + 1x_1}{k+1}$$

$$\frac{-3}{k-1} = \frac{k(0) - 1(6)}{k-1} \quad \checkmark$$

$$-3k + 3 = -6$$

$$-3k = -9$$

$$k = 3 \quad \checkmark$$

1 for substituting correctly

1 for value for k .

c) $\left| \frac{5}{x} + 3 \right| > 3$

1 for each case & solution (2 marks)

note: $x \neq 0$

1 for overall solution

$$\frac{5}{x} + 3 > 3 \quad \text{or} \quad \frac{5}{x} + 3 < -3$$

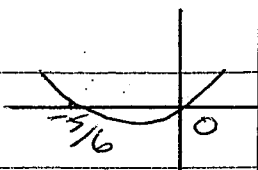
$$\frac{5}{x} > 0 \quad \frac{5}{x} < -6$$

$$5x > 0 \quad 5x < -6x^2$$

$$x > 0 \quad \checkmark \quad 6x^2 + 5x < 0$$

$$x(6x+5) < 0$$

$$-\frac{5}{6} < x < 0 \quad \checkmark$$



∴ overall solution is $x > -\frac{5}{6}, x \neq 0 \quad \checkmark$

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$$\text{OR } \left| \frac{5}{x} + 3 \right| > 3$$

note: $x \neq 0$

$$\text{when } \frac{5}{x} + 3 > 3 \quad \text{or} \quad \frac{5}{x} + 3 < -3$$

$$\frac{5}{x} > 0$$

$$\frac{5}{x} < -6$$

$$\therefore x > 0 \quad \checkmark$$

$$\text{If } x < 0,$$

$$5 > -6x$$

$$-\frac{5}{6} < x$$

$$\text{i.e. } -\frac{5}{6} < x < 0 \quad \checkmark$$

$$\therefore \text{overall solution is } x > -\frac{5}{6}, x \neq 0 \quad \checkmark$$

$$d) f(x) = (x^2 - 16)Q(x) + (3x - 1) \quad \checkmark \quad \leftarrow$$

$$f(4) = (4^2 - 16)Q(4) + (12 - 1)$$

$$= 0 + 11$$

$$= 11$$

$$\therefore \text{remainder is } 11. \quad \checkmark$$

1 mark for
expressing $f(x)$
in terms of
(divisor)(quotient)
+ remainder

1 mark for
remainder.

a) $\sin 105^\circ = \sin(60^\circ + 45^\circ)$

$$= \sin 60^\circ \cos 45^\circ + \cos 60^\circ \sin 45^\circ \checkmark$$

$$= \left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{2} \times \frac{1}{\sqrt{2}}\right)$$

$$= \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

OR

$$= \frac{\sqrt{6} + \sqrt{2}}{4} \checkmark$$

1 mark for expansion

1 mark for exact value

b) $\tan \theta = 3 \cot \theta$

$$\tan^2 \theta = 3$$

$$\tan \theta = \sqrt{3}, \tan \theta = -\sqrt{3} \checkmark$$

$$\therefore \theta = n\pi + \frac{\pi}{3} \text{ or } \theta = n\pi + \frac{2\pi}{3} \checkmark$$

1 mark for

obtaining $\tan \theta = \pm \sqrt{3}$

1 mark for both general solutions

c) $\cos(A+B) + \cos(A-B)$

$$= \cos A \cos B - \sin A \sin B + \cos A \cos B + \sin A \sin B \checkmark$$

$$= 2 \cos A \cos B$$

1 mark for expanding

$$\therefore \int_0^{\pi/6} 2 \cos 3x \cos 2x \, dx$$

$$= \int_0^{\pi/6} (\cos 5x + \cos x) \, dx$$

$$= \left[\frac{1}{5} \sin 5x + \sin x \right]_0^{\pi/6} \checkmark$$

$$= \left[\frac{1}{5} \sin \frac{5\pi}{6} + \sin \frac{\pi}{6} \right] - 0$$

$$= \frac{1}{5} \times \frac{1}{2} + \frac{1}{2} = \frac{3}{5} \checkmark$$

1 mark for

primitive function

1 mark for value

$$d) \sin x = \frac{2t}{1+t^2}$$

1 mark for expressing $\sin x$ & $\cos x$ in terms of t

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$5 \cos x - 2 \sin x = 2$$

$$5 \left(\frac{1-t^2}{1+t^2} \right) - 2 \left(\frac{2t}{1+t^2} \right) = 2$$

$$5 - 5t^2 - 4t = 2 + 2t^2 \quad \checkmark$$

$$7t^2 + 4t - 3 = 0$$

$$(7t-3)(t+1) = 0$$

$$\therefore t = \frac{3}{7} \quad \text{or} \quad t = -1$$

$$\tan \frac{x}{2} = \frac{3}{7}, \quad \tan \frac{x}{2} = -1 \quad \checkmark \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{1 mark here}$$

$$\text{Since } 0 \leq x \leq 2\pi, \quad 0 \leq \frac{x}{2} \leq \pi$$

$$\frac{x}{2} \approx 0.4049, \quad \frac{3\pi}{4}$$

$$\therefore x = 0.8098, \quad \frac{3\pi}{2} \quad \checkmark$$

1 mark for both correct solutions for x

$$a) \lim_{\theta \rightarrow 0} \frac{\theta + \sin 2\theta}{3\theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{1}{3} + \frac{2 \sin \theta \cos \theta}{3\theta} \quad \checkmark$$

$$= \frac{1}{3} + \frac{2}{3} \times 1 \times 1 \quad \checkmark$$

$$= 1 \quad \checkmark$$

OR

$$\lim_{\theta \rightarrow 0} \frac{\theta + \sin 2\theta}{3\theta}$$

$$= \lim_{\theta \rightarrow 0} \frac{1}{3} + \frac{\sin 2\theta}{2\theta} \times \frac{2}{3} \quad \checkmark$$

$$= \frac{1}{3} + 1 \times \frac{2}{3} \quad \checkmark$$

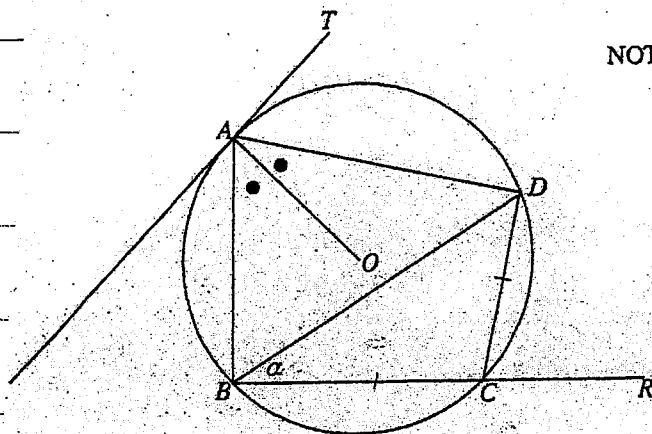
$$= 1 \quad \checkmark$$

1 mark for deducing
how to use fundamental
limit to help evaluate

} 2 marks for working

b)

NOT TO SCALE



i) 1 mark for
saying ext. $\angle$ of $\triangle DCB$

ii) 1 mark for
realising $\angle BAD = \angle DCR$
+

1 mark for reason
for this

iii) marks shown
ever.

$\angle BDC = \alpha$ (base $\angle$ s of isosceles $\triangle DCB$)

$\therefore \angle DCR = 2\alpha$ (exterior $\angle$ of $\triangle DCB$)

} $\checkmark$

b) ii) $\angle BAD = \angle DCR = 2\alpha$ ✓

(exterior $\angle$ of cyclic quad^l ABCD ✓
equals interior opposite $\angle$)

$\therefore \angle OAD = \alpha$ (given OA bisects $\angle BAD$)

iii) $\angle TAD = 90^\circ - \alpha$

(radius $\perp$ tangent at pt. of contact) } ✓ 1 mark

$\angle ABD = 90^\circ - \alpha$

($\angle$ in alternate segment)

$\therefore \angle ABC = 90^\circ - \alpha + \alpha = 90^\circ$ ✓

} 1 mark.

a) $P(x) = x^3 - 6x^2 + ax - 4, a > 0$

roots: $\alpha, \beta, \alpha\beta$ ^{real & positive!}

$$\alpha + \beta + \alpha\beta = 6 \quad (1)$$

$$\alpha\beta + \alpha\beta^2 + \alpha^2\beta = \alpha\beta(1 + \beta + \alpha) = a \quad (2)$$

1 mark.

$$(\alpha\beta)^2 = 4 \quad (3)$$

$\therefore \alpha\beta = 2$ sub into (1) $\leftarrow$ 1 mark for $\alpha\beta = 2$

$\therefore \alpha + \beta = 4$ sub into (2) $\leftarrow$ 1 mark for $\alpha + \beta = 4$

$$2(1 + 4) = a$$

$$\therefore a = 10.$$

1 mark for finding 'a'.

b) i) $y = \frac{x^2}{4}$

$$y' = \frac{x}{2} \text{ at } x = 2a,$$

$$y' = a \text{ (m of tangent at P)}$$

1 mark for saying tangent at P has $m = a$ & using $m = \tan \theta$.

Now $m = \tan \theta$

$$a = \tan \theta.$$

ii) $m_{SP} = \frac{a^2 - 1}{2a}$

$$= \frac{1}{2} \left(\frac{a^2}{a} - \frac{1}{a} \right)$$

$$= \frac{1}{2} \left(a - \frac{1}{a} \right)$$

1 mark for using gradient formula & rearranging to obtain answer given.

$$\text{iii) } \tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

1 mark for substituting
into formula
1 mark for
working that leads
to required answer

$$= \frac{a - \frac{1}{2}(a - \frac{1}{a})}{1 + a \cdot \frac{1}{2}(a - \frac{1}{a})}$$

✓

$$= \frac{a - \frac{1}{2}a + \frac{1}{2a}}{1 + \frac{a^2}{2} - \frac{1}{2}}$$

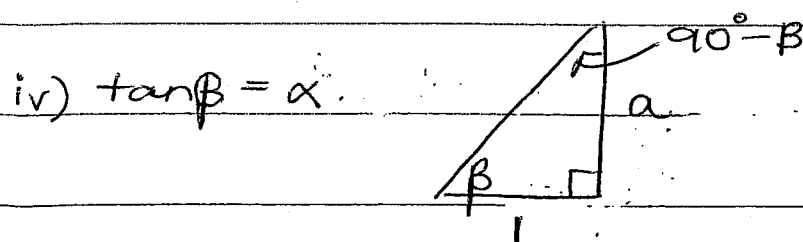
$$= \frac{\frac{a}{2} + \frac{1}{2a}}{\frac{1}{2} + \frac{a^2}{2}}$$

$$\tan \theta = \frac{\frac{a^2 + 1}{2a} \div \frac{1 + a^2}{2}}$$

$$= \frac{\frac{2}{2a}}$$

$$\tan \theta = \frac{1}{a}$$

✓



$$\therefore \cot(90^\circ - \beta) = \frac{1}{a} = \tan \theta \quad \checkmark \leftarrow 1 \text{ mark}$$

$$\therefore 90^\circ - \beta = \theta \quad \checkmark \leftarrow 1 \text{ mark.}$$

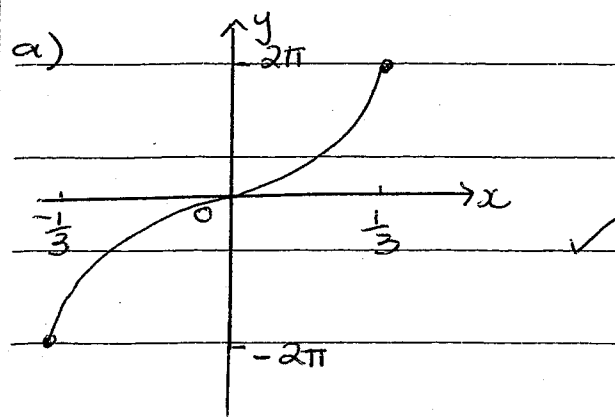
$$\therefore \theta + \beta = 90^\circ$$

v) If $\theta = \beta$

$$\theta + \beta = 2\beta = 90^\circ \therefore \beta = 45^\circ \quad \checkmark \leftarrow 1 \text{ mark}$$

$$\tan 45^\circ = a \therefore a = 1$$

$$\therefore P \text{ is } (2, 1) \quad \checkmark \leftarrow 1 \text{ mark}$$

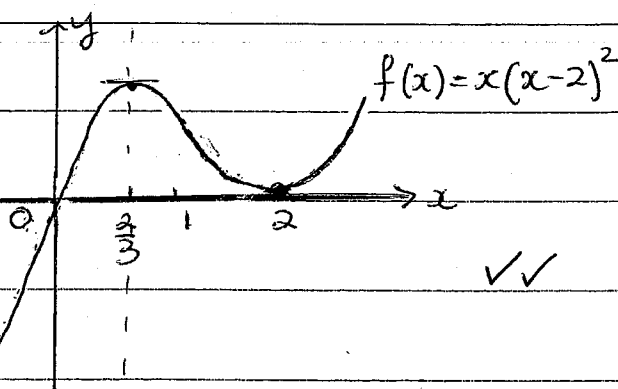


1 mark for graph
1 mark for both
domain & range.

D: $-\frac{1}{3} \leq x \leq \frac{1}{3}$ ✓

R: $-2\pi \leq y \leq 2\pi$

b) i)



1 mark for shape

1 mark for showing
y-intercept, where
turning pts. are

ii) $a = \frac{2}{3}$ ✓. If $x \leq \frac{2}{3}$,

$f(x)$ is a 1:1 fn ∴ its
inverse will exist. ✓

1 mark for 'a'.
1 mark for reason

iii) If $x = \frac{2}{3}$, $y = \frac{5}{27}$

D: $x \leq \frac{2}{3}$ $R_f: y \leq \frac{5}{27}$

∴ D: $x \leq \frac{5}{27}$ ✓ ← 1 mark.
f⁻¹



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$$a) \sin^{-1} x - \cos^{-1} y = \frac{\pi}{12} \quad (1)$$

$$\sin^{-1} y + \cos^{-1} x = \frac{5\pi}{12} \quad (2)$$

$$\sin^{-1} x + \cos^{-1} x - \cos^{-1} y + \sin^{-1} y \quad (1) + (2)$$

$$\frac{\pi}{2} - \cos^{-1} y + \sin^{-1} y = \frac{\pi}{2} \quad \checkmark \quad 1 \text{ mark}$$

$$\sin^{-1} y = \cos^{-1} y$$

$$\text{when } y = \frac{1}{\sqrt{2}} \text{ sub in (1)} \quad \checkmark \quad 1 \text{ mark for 'y'}$$

$$\sin^{-1} x - \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{12}$$

$$\sin^{-1} x = \frac{\pi}{12} + \frac{\pi}{4} = \frac{\pi}{3}$$

$$\therefore x = \frac{\sqrt{3}}{2} \quad \checkmark$$

1 mark for 'x'.

$$\therefore \text{the soln is } \left(\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}\right)$$

a) $\int \frac{1}{\sqrt{4-x^2}} dx$

$= \sin^{-1} \frac{x}{2} + C \checkmark$

← 1 mark. Do not take mark off if no 'C'

b) $y = \cos^{-1}(3-2x)$ let $u = 3-2x$

$y' = \frac{dy}{du} \times \frac{du}{dx}$

$= -\frac{1}{\sqrt{1-(3-2x)^2}} \times -2$

← 1 mark

$\checkmark$

$= \frac{2}{\sqrt{(1+3-2x)(1-(3-2x))}}$

$= \frac{2}{\sqrt{(4-2x)(2x-2)}}$

$= \frac{2}{\sqrt{4(2-x)(x-1)}}$

$= \frac{1}{\sqrt{(2-x)(x-1)}}$

← 1 mark for factorising denominator

$\checkmark$

$\checkmark$

← 1 mark for simplest factored form.

c) $\int_0^{\frac{1}{\sqrt{3}}} \frac{dx}{1+9x^2}$

$= \frac{1}{9} \int_0^{\frac{1}{\sqrt{3}}} \frac{dx}{(\frac{1}{9} + x^2)}$

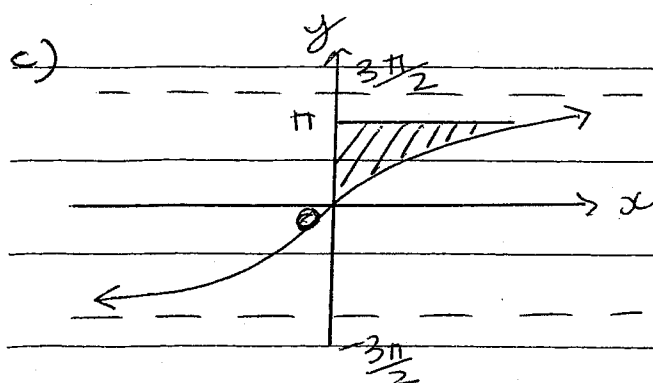
$= \frac{1}{9} \left[3 \tan^{-1} 3x \right]_0^{\frac{1}{\sqrt{3}}}$

$= \frac{1}{9} \left[3 \tan^{-1} \frac{3}{\sqrt{3}} - 0 \right]$

$= \frac{1}{9} \left[3 \tan^{-1} \sqrt{3} \right] = \frac{1}{3} \times \frac{\pi}{3} = \frac{\pi}{9} \checkmark$

← 1 mark for primitive fn

← 1 mark for value exact



$$y = 3 \tan^{-1} x$$

$$x = \tan \frac{y}{3} \quad \checkmark$$

1 mark for making 'x' the subject

$$\therefore V = \pi \int_0^{\pi} \tan^2 \frac{y}{3} dy$$

$$= \pi \int_0^{\pi} (\sec^2 \frac{y}{3} - 1) dy \quad \checkmark$$

1 mark for writing in terms of $\sec^2 \theta$.

$$= \pi \left[3 \tan \frac{y}{3} - y \right]_0^{\pi} \quad \checkmark$$

1 mark for integrating correctly

$$= \pi \left[3 \tan \frac{\pi}{3} - \pi - 0 \right]$$

$$= \pi \left[3\sqrt{3} - \pi \right] \text{ units}^3 \quad \checkmark$$

1 mark for exact volume

